

Coefficient residues in generalized Fibonacci products

Finite-state and signed-cancellation proofs of two conjectures of Stanley

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Abstract

For the k -bonacci numbers initialized by k ones, consider the product $P_n(x) = \prod_{i=0}^{n-1} (1 + x^{F_{i+k}^{(k)}})$, with $k \geq 2$. We give a self-contained finite-state argument that, for every modulus m and residue a , the number of coefficients of P_n congruent to a has a rational ordinary generating function in n . The construction also proves rationality for the number of coefficients equal to any prescribed positive integer. Its key ingredients are a unique binary normal form avoiding $1^k 0$, a factorization of each representation fiber into disjoint carry blocks, and an exact counter with only $k+1$ coordinates. Modulo two, a uniform automaton with $4k+2$ states gives

$$\sum_{n \geq 0} \#\{j : [x^j]P_n(x) \text{ is odd}\} z^n = \frac{1 + 2z^k}{1 - 2z + 2z^k - 2z^{k+1}}.$$

We then prove that same odd-coefficient identity a *second* time, by a completely disjoint elementary route: replacing selected plus signs by minus signs according to a periodic coherent signing makes every coefficient of the signed product lie in $\{-1, 0, 1\}$, parity becomes a squared norm, and an exact overlap identity closes the recurrence. The second proof shares no lemma with the first, and it holds for every quasifibonacci weight sequence with superincreasing seeds, not only for Stanley's initialization. These arguments provide proposed proofs of Conjectures 6.2 and 6.3 in Stanley's arXiv version 3. We explicitly handle internal zero coefficients, prove two *different* exponential sparsity statements with different denominators, derive exponential asymptotics twice (analytically and by Perron–Frobenius), and include two independent exact computational suites and finite certificates for two modulo-three identities. None of the general theorems depends on a finite numerical fit.

Status and scope. This is an unrefereed research manuscript containing complete proposed proofs, not a claim of independently certified discovery. The identified source explicitly states the target conjectures. A targeted literature search did not identify a later resolution, but does not establish priority. All claims below are restricted to $k \geq 2$ and to the stated initial conditions, except where a statement is explicitly extended to arbitrary superincreasing seeds.

Companion artifacts. Editable L^AT_EX source; two independent standard-library Python implementations; two exact verification suites and a cross-verification script that relates them; CSV tables; JSON automata and Cayley–Hamilton certificates. The symbolic cross-check additionally uses SymPy.

Keywords. Generalized Fibonacci numbers; quasifibonacci partitions; coefficient congruences; flat signed products; weighted automata; rational generating functions; binary normalization.

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1 The problem and the main results

An exponentially large polynomial can contain substantial finite-state structure. The issue is to expose that structure without confusing representations of an exponent with distinct exponents, or confusing the number of representations modulo m with the number of exponents having that residue. Both distinctions matter in the problem considered here.

A second, quite different mechanism is also available for the modulus two: a suitable periodic signing of the factors makes the product *flat*, and parity then becomes a squared norm. This manuscript carries both mechanisms, because they prove the same identity from disjoint hypotheses and each reaches conclusions the other cannot.

1.1 The documented target

Stanley’s paper *Theorems and Conjectures on Some Rational Generating Functions* asks whether coefficient-residue counts for generalized Fibonacci products always have rational generating functions. It also proposes a particularly small formula for the odd-coefficient count [1, Conjectures 6.2 and 6.3, pp. 23–24]. Our numbering of those conjectures refers specifically to arXiv:2101.02131v3. The article subsequently appeared in the *European Journal of Combinatorics* in 2024 [2].

The relevant background includes Stanley’s factorization of equal-weight pairs [1, Lemma 5.2] and the symbolic dynamic programming methods of Ekhad and Zeilberger [6]. The finite-state argument below instead organizes one canonical representative per exponent and counts its fiber. We prove the required canonical-pair factorization directly, so that central argument has no dependence on an omitted proof from the literature.

For the second proof there is also important prior work to acknowledge. Zhao proved flatness of finite signed Fibonacci products when $k = 2$ [4]. Diao introduced the quasifibonacci seed conditions used in Section 9 and proved flatness of the infinite all-minus product for even recurrence order [3, Definition 1.1 and Theorem 1.1]. The argument here uses a varying, periodic signing, works for both even and odd orders, concerns every finite product, and gives its exact support count. These distinctions matter: simply replacing every plus sign by a minus sign does *not* work for odd k (Section 9.7).

1.2 Provenance of this manuscript

This article is the merger of two previously separate research packages that independently attacked the same conjecture page.

- **stanley-kbonacci-residues** (September 20, 2026) proved Conjecture 6.2 for every modulus and residue, and proved Conjecture 6.3 by constructing a unique regular cross-section of the representation map, a proved factorization of each canonical fiber into carry blocks, an exact $(k + 1)$ -coordinate counter, a uniform $4k + 2$ -state parity automaton, and a symbolic elimination valid for arbitrary k . It is the structural spine of the present article: Sections 2–8, the combinatorial half of Section 11, and Sections 13–14 come from it.
- **quasifibonacci-odd-coefficient-counts** (September 19, 2026) proved Conjecture 6.3 by a signed-cancellation argument: a coherent periodic signing, separation inequalities for the weights, a signed reflection identity, a pointwise two-tail cancellation identity, flatness, and an exact overlap inner product. It proved the statement for *every* superincreasing seed list, and it explicitly recorded that its parity lift cannot reach moduli greater than two. Sections 9–10, the analytic half of Section 11, Section 12, and Appendix A come from it.

[Kept twice: literature audits.] Each package carried its own literature and status audit, and neither list contains the other. Both are retained: `LITERATURE.md` records which printed pages of the source were inspected as rendered images, Stanley’s Lemma 5.2 background, and the Ekhad–Zeilberger version dating; `RESEARCH_STATUS.md` records Zhao, Diao, OEIS A104767 with the exact relation $a_2(n) = 2A104767(n)$ for $n \geq 1$, and Tang–Xin [7]. The bibliography below is their union.

The two packages shared no lemma. Their agreement on the numbers was checked before the merge (for instance $a_3(4) = 14$ in both, and the growth constant $\rho_2 = 1.5436890126920763615\dots$ from the 80-digit bisection against 1.5436890126920766 from the independent floating-point bisection), and it is now checked automatically by `code/cross_check.py` (Section 13.2).

1.3 Definitions and conventions

Fix an integer $k \geq 2$. Define

$$F_1^{(k)} = \dots = F_k^{(k)} = 1, \quad F_{i+1}^{(k)} = \sum_{j=0}^{k-1} F_{i-j}^{(k)} \quad (i \geq k). \quad (1.1)$$

Set $w_i = F_{i+k}^{(k)}$ for $i \geq 0$. Thus the first weights are

$$\begin{array}{c|cccccccc} k=2 & 1 & 2 & 3 & 5 & 8 & 13 & 21 & 34 \\ k=3 & 1 & 3 & 5 & 9 & 17 & 31 & 57 & 105 \\ k=4 & 1 & 4 & 7 & 13 & 25 & 49 & 94 & 181. \end{array}$$

In particular, the $k = 3$ initialization is *not* the alternative 1, 2, 4, 7, $\dots$ sometimes used for tribonacci numeration.

Write

$$P_n(x) = \prod_{i=0}^{n-1} (1 + x^{w_i}) = \sum_{j=0}^{D_n} c_n(j) x^j, \quad D_n = \sum_{i=0}^{n-1} w_i, \quad (1.2)$$

with $P_0 = 1$ and $D_0 = 0$. All the $c_n(j)$ are nonnegative integers. For $m \geq 2$ and $0 \leq a < m$, distinguish

$$h_{m,a}^{(k)}(n) = \#\{j : 0 \leq j \leq D_n, c_n(j) > 0, c_n(j) \equiv a \pmod{m}\}, \quad (1.3)$$

$$\mathfrak{h}_{m,a}^{(k)}(n) = \#\{j : 0 \leq j \leq D_n, c_n(j) \equiv a \pmod{m}\}. \quad (1.4)$$

Their ordinary generating functions are denoted $H_{m,a}^{(k)}(z)$ and $\mathcal{H}_{m,a}^{(k)}(z)$, respectively. The first convention counts nonzero terms; the second counts every coefficient position through the degree. They differ only at residue zero. Proving both avoids any dependence on how the word “coefficients” is interpreted. Coefficients at exponents beyond the degree are never counted. This is a finite-product problem: n is the number of factors, not an upper bound on the exponent being counted.

We abbreviate the odd count by

$$a_k(n) = h_{2,1}^{(k)}(n) = \mathfrak{h}_{2,1}^{(k)}(n), \quad a_k(0) = 1, \quad (1.5)$$

the two conventions agreeing at residue one because a missing position has coefficient zero, which is even.

For a word $u = u_0 \cdots u_{n-1} \in \{0, 1\}^n$, define

$$\text{val}(u) = \sum_{i=0}^{n-1} u_i w_i. \quad (1.6)$$

Throughout the article, words are written and read in *increasing weight order*: the leftmost digit has weight w_0 . Expansion of (1.2) shows that $c_n(j)$ is exactly the number of length- n words with value j .

1.4 The main results

Theorem 1.1 (Residue rationality; Conjecture 6.2). *For all $k \geq 2$, $m \geq 2$, and $0 \leq a < m$, both $H_{m,a}^{(k)}(z)$ and $\mathcal{H}_{m,a}^{(k)}(z)$ belong to $\mathbb{Q}(z)$. The nonzero-term function has an explicitly constructible integer linear representation with at most $(k+1)m^{k+1}$ states.*

Theorem 1.2 (The odd-coefficient formula; Conjecture 6.3). *For every $k \geq 2$,*

$$H_{2,1}^{(k)}(z) = \mathcal{H}_{2,1}^{(k)}(z) = \sum_{n \geq 0} a_k(n) z^n = \frac{1 + 2z^k}{1 - 2z + 2z^k - 2z^{k+1}}. \quad (1.7)$$

Theorem 1.3 (Prescribed exact multiplicity). *Fix $k \geq 2$ and $q \geq 1$. Then*

$$\sum_{n \geq 0} \#\{j : c_n(j) = q\} z^n$$

is rational. A finite-state construction uses at most $(k+1)(q+2)^{k+1}$ states.

Theorem 1.4 (Seed independence and signed flatness). *Let $w_0, w_1, \dots$ be any quasifibonacci sequence of order k in the sense of Definition 9.1: positive integers with superincreasing seeds $w_i > \sum_{j < i} w_j$ for $0 \leq i \leq k-1$, continued by the k -term recurrence. Let $\varepsilon_0, \varepsilon_1, \dots$ be any coherent signing (Definition 9.2). Then every coefficient of the signed product $\prod_{i < n} (1 + \varepsilon_i x^{w_i})$ lies in $\{-1, 0, 1\}$; the parity of $[x^n] \prod_{i < n} (1 + x^{w_i})$ is the square of the corresponding signed coefficient; and the number of odd coefficients is independent of the seeds and of the signing and again equals $a_k(n)$ in (1.7).*

Theorems 1.1–1.3 are proved by the finite-state route of Sections 2–8. Theorem 1.4 is proved by the signed route of Section 9, which *re-proves* Theorem 1.2 as the special case of Stanley’s seeds (Lemma 9.6). The two proofs of (1.7) are logically independent; see Section 1.5.

1.5 What this article deliberately keeps twice

Merging two disjoint solutions of one problem is only worthwhile if the duplication is preserved where it carries information. The following thirteen items are kept in duplicate on purpose. Each is flagged in the text with the marker **[Kept twice: <item>]**.

- D1. Conjecture 6.3 itself**, proved once in Sections 2–8 (canonical words, fiber factorization, counter, $4k+2$ -state automaton, symbolic elimination) and once in Section 9 (coherent signing, separation, reflection, two-tail identity, parity lift, norm recurrence). *What it buys*: two arguments with no lemma in common confirm each other, so an error in either route cannot silently survive.
- D2. Two structural witnesses for the same coefficients**: flatness gives the parity as $\varphi_n(m)^2$ with $\varphi_n(m) \in \{-1, 0, 1\}$ (Theorem 9.3), while the counter gives the exact integer $c_n(j)$ (Proposition 4.1). *What it buys*: neither implies the other, and their agreement — $c_n(j)$ odd exactly when $|\varphi_n(j)| = 1$ — is a genuine cross-check (Section 13.2).
- D3. The dominant-root asymptotic** $a_k(n) = \gamma_k \rho_k^n + O(\theta_k^n)$, proved analytically in Section 11.1 and combinatorially by Perron–Frobenius in Section 11.2. *What it buys*: the analytic proof needs no automaton and the combinatorial proof needs no complex analysis, so the asymptotic survives if either technique is questioned. The two explicit leading constants C_k and γ_k are both kept, with their equality proved in Corollary 11.3.
- D4. The inequality** $\rho_k < \beta_k$ (odd growth below k -bonacci growth), proved once from $s^k < 1/2$ and $L_k(s) < 1$ (Section 11.3) and once from $V_k(t_k) = t_k^k(2 - 3t_k) > 0$ (Theorem 11.2). *What it buys*: the strict separation that makes both sparsity statements meaningful is independent of which growth polynomial one prefers.
- D5. Two different sparsity statements**, not one repeated: density among *all* positions $0 \leq j \leq D_n$ including zeros (Theorem 11.4) and density among *nonzero* coefficients only (Theorem 11.2). *What it buys*: the first was explicitly disclaimed by the signed route and the second is inaccessible to it, so stating only one would lose a theorem.
- D6. Two numerical growth-constant computations**: an 80-digit Decimal bisection (data/growth.csv, Table 3) and an independent floating-point bisection (data/growth_constants.csv, Table 4). *What it buys*: agreement to floating-point precision between two independently coded root finders, with the 80-digit values as the primary record.

- D7. Two count tables of the same sequence:** `data/counts.csv` ($2 \leq k \leq 20$, $n \leq 200$) and `data/odd_counts.csv` ($2 \leq k \leq 8$, $n \leq 100$), produced by different algorithms. *What it buys:* equality on the overlap is an independent check of both proofs at once, not of one implementation against itself.
- D8. Two algorithm families for the odd count:** the order- $(k+1)$ recurrence and $X^N \bmod \chi_k$ polynomial powering on one side; direct expansion, the generic mod-2 machine, and the uniform $4k+2$ -state table on the other (Section 12). *What it buys:* an independent check, because the two families descend from different proofs — whereas two recurrence-based routines agreeing would check only implementation.
- D9. Two coefficient oracles with different outputs:** `CoefficientOracle.signed` returns the signed value in $\{-1, 0, 1\}$ for arbitrary superincreasing seeds and arbitrary initial sign blocks, and `coefficient_for_canonical` returns the exact nonnegative integer coefficient for Stanley’s seeds. *What it buys:* neither subsumes the other, and the assertion $|\text{signed}| \equiv \text{coefficient} \pmod{2}$ links them.
- D10. Two exhaustive verification suites and their two coverage tables** (Tables 5 and 6), testing disjoint objects. *What it buys:* the record of what was actually checked; collapsing them into one summary would destroy it.
- D11. Two proof-hygiene mechanisms:** the dependency map of Appendix A, which records that no rational-generating-function ansatz and no unproved word-rewriting monoid is assumed; and the Cayley–Hamilton certificate Lemma 13.1, which records that a proved state dimension, not an arbitrary cutoff, makes finite checks sufficient. *What it buys:* the two articles’ respective answers to “why is this not curve fitting.”
- D12. Two initialization discussions:** Remark 9.13 (why the superincreasing hypothesis cannot be dropped) and Section 14.2 (why $k=1$ is excluded). *What it buys:* they rule out different degenerations, one in the seeds and one in the order.
- D13. Two literature audits:** `RESEARCH_STATUS.md` (Zhao, Diao, OEIS A104767 with $a_2(n) = 2A104767(n)$ for $n \geq 1$, Tang–Xin) and `LITERATURE.md` (Stanley’s Lemma 5.2 background, the page-image inspection record, the Ekhad–Zeilberger dating). *What it buys:* neither list contains the other, and the merged bibliography is their union.

1.6 Notation reconciled between the two routes

The two source packages used incompatible symbols. This article uses one convention throughout; the table records the translation, so that readers of either original can navigate.

This article	Finite-state package	Signed package
$w_0, \dots, w_{n-1}, w_i = F_{i+k}^{(k)}$	same	$w_1, \dots, w_n, w_i = F_{i+k-1}^{(k)}$
$D_n = \sum_{i < n} w_i$ (degree)	D_n	S_n
$P_n(x), c_n(j)$	same	$I_n(x)$
$a_k(n)$ (odd count)	$a_k(n) = H_{2,1}^{(k)}$	$h_k(n)$
$\Phi_n(x), \varphi_n(m)$ (signed product)	—	$A_n(x), a_n(m)$
$\Pi_n = \prod_{i < n} \varepsilon_i$	—	E_n
G_n (overlap inner product)	—	B_n
$V_k(z) = 1 - 2z + 2z^k - 2z^{k+1}$	$V_k(z)$	$D_k(z)$
$R_k(z) = 1 + z + \dots + z^{k-1}$	R	$B_k(z)$
$B_k(z) = \sum_{j=2}^{k-1} (k-j)z^j$ (holes)	$B_k(z)$	—
$L_k(z) = z + \dots + z^{k-1} + 2z^{2k}$	—	$L_k(z)$
r_k (root in $(0, 1)$)	r_k	r_k (same number)
$\rho_k = 1/r_k$ (odd growth)	ρ_k	λ_k
β_k (k -bonacci growth)	β_k	ρ_k
θ_k (subdominant bound)	θ_k	μ_k

Three symbols are reused with different meanings in disjoint scopes, and we flag them once here rather than inventing new letters.

- In Sections 7–8 only, the letters $A_j, B_j, C_j, D, E_j, F, X$ denote automaton states and, in Section 8, the generating series of accepted continuations from those states. Elsewhere $B_k(z)$ is the hole numerator, D_n is the degree, and C_k is the leading asymptotic constant of Theorem 11.1. No formula mixes the two readings.
- $R_k(z)$ is the polynomial $1 + z + \dots + z^{k-1}$ throughout. It is what the signed package called $B_k(z)$; the symbol $B_k(z)$ in this article always means the hole numerator $\sum_{j=2}^{k-1} (k-j)z^j$ of (5.3).
- The supplied Python module `kbonacci_parity.py` stores weights and signs in one-based internal arrays, matching the signed package's original indexing. Its public functions take and return the same weight *sets* as `kbonacci_residues.py`; only the internal offset differs, and `code/cross_check.py` asserts the agreement.

2 Unique normal forms for represented exponents

The weight sequence satisfies

$$w_0 = 1, \quad w_i = (k-1)2^{i-1} + 1 \quad (1 \leq i < k), \quad w_i = \sum_{j=1}^k w_{i-j} \quad (i \geq k). \quad (2.1)$$

The middle identity follows by induction from (1.1); it records the exceptional initial segment that must not be replaced by a generic k -bonacci initialization. Section 9 works with a much larger class of initial segments, but the normal form of the present section is tied to (2.1).

Definition 2.1. A binary word is *canonical* if it avoids the consecutive substring $1^k 0$. Let $\mathcal{L}_k$ be the language of canonical words, including the empty word. A normalization step is the replacement

$$1^k 0 \longrightarrow 0^k 1 \quad (2.2)$$

inside a fixed-length word.

Every normalization step preserves value, by the recurrence for w_i . It strictly increases the ordinary binary integer $\sum_i u_i 2^i$: a step starting at position r increases that integer by 2^r . There are only finitely many words of a fixed length. Therefore every sequence of normalization steps terminates, and its terminal word is canonical. Uniqueness requires a separate argument.

Lemma 2.2 (A lower-weight bound). *If a word u of length i has no k consecutive ones, then $\text{val}(u) < w_i$.*

Proof. For $i = 0$ the assertion is $0 < 1$. For $1 \leq i < k$, the sum of all the lower weights is less than w_i ; indeed (2.1) gives

$$w_i - \sum_{j=0}^{i-1} w_j = k - i > 0. \quad (2.3)$$

Proceed by induction for $i \geq k$. Let r be the run of ones at the high-weight end of u . Then $0 \leq r \leq k-1$. The next digit is zero, and the remaining word has length $i - r - 1$. The induction hypothesis gives

$$\text{val}(u) < w_{i-1} + \dots + w_{i-r} + w_{i-r-1} \leq w_i.$$

Here the first sum is empty if $r = 0$; the final inequality holds because these are at most the k most recent positive weights in the recurrence. $\square$

Equation (2.3) is exactly the statement that Stanley's own seeds are superincreasing; it is reused in Lemma 9.6 to make Theorem 1.4 imply Theorem 1.2.

Lemma 2.3 (Canonical injectivity). *Two distinct canonical words of the same length have distinct values.*

Proof. First observe that every portion of a canonical word strictly below a zero digit contains no k consecutive ones. Otherwise a run of at least k ones would eventually meet a zero, producing the forbidden pattern.

Now take two distinct canonical words and let i be their highest position of disagreement. Suppose the first has a one there and the second a zero. Their higher contributions agree. By the observation and Lemma 2.2, the entire lower contribution of the second word is less than w_i . The first word contributes w_i at position i and a nonnegative amount below it. Its value is therefore strictly larger. $\square$

Proposition 2.4 (A regular cross-section). *Every fiber of $\text{val} : \{0, 1\}^n \rightarrow \mathbb{N}$ contains exactly one canonical word. Thus normalization terminates at the same word regardless of the order of the steps. Moreover,*

$$\mathcal{L}_k = (0 \mid 10 \mid \cdots \mid 1^{k-1}0)^* 1^*. \quad (2.4)$$

Proof. Termination supplies a canonical word in every fiber, and Lemma 2.3 gives uniqueness. In a canonical word, the run of ones before each zero has length less than k ; the final run of ones is unrestricted. Cutting immediately after each zero gives exactly (2.4), with unique cuts. $\square$

This avoids a potentially invalid shortcut: termination by itself does not establish uniqueness of a rewriting system. Here uniqueness is a consequence of an explicit numerical separation argument.

Corollary 2.5 (Support size). *Let $s_k(n) = \#\{j : c_n(j) > 0\}$. Then*

$$S_k(z) := \sum_{n \geq 0} s_k(n) z^n = \frac{1}{(1-z)(1-z-z^2-\cdots-z^k)} = \frac{1}{1-2z+z^{k+1}}. \quad (2.5)$$

Proof. Canonical words are in bijection with represented exponents. In (2.4), a block ending in zero has generating function $z + \cdots + z^k$, and the final run of ones has generating function $1/(1-z)$. Unique decomposition gives the first fraction; multiplication of its denominator gives the second. $\square$

The support generating function (2.5) is available only on this route; the signed argument of Section 9 has no access to it. That is precisely why the two sparsity statements of Section 11 are different theorems.

3 The structure of a canonical representation fiber

Counting canonical words counts exponents, but does not yet reveal their coefficients. We need to count *all* words with the value of a given canonical word. The next factorization turns that task into a small weighted automaton.

For $j \geq 0$ and $c_1, \dots, c_j \in \{0, 1\}$, call

$$\begin{pmatrix} 0^k(c_1 0^{k-1}) \cdots (c_j 0^{k-1}) 1 \\ 1^k(c_1 1^{k-1}) \cdots (c_j 1^{k-1}) 0 \end{pmatrix} \quad (3.1)$$

a *carry block*. Parentheses indicate concatenation blocks, not multiplication. When $j = 0$, this is simply the pair $(0^k 1, 1^k 0)$.

Theorem 3.1 (Canonical-fiber factorization). *Let a be canonical and let b be any binary word of the same length. Then $\text{val}(a) = \text{val}(b)$ if and only if the two-row word (a, b) is a concatenation of diagonal columns $(0, 0)$ and $(1, 1)$ and carry blocks (3.1). Such a factorization, when it exists, is unique.*

Proof. Necessity. Normalize b to a , using Proposition 2.4. For each integer i , let t_i be the number of normalization steps that start at position i . Set $t_i = 0$ outside $0 \leq i \leq n - k - 1$. Summing the changes at each coordinate gives

$$a_i = b_i - \sum_{h=0}^{k-1} t_{i-h} + t_{i-k} \quad (0 \leq i < n). \quad (3.2)$$

All t_i are nonnegative integers. We use only this equation and the binary bounds on a_i, b_i .

If every t_i is zero, the rows agree. Otherwise let r be the least index with $t_r > 0$. At position r , equation (3.2) reduces to $a_r = b_r - t_r$. Hence

$$t_r = 1, \quad (a_r, b_r) = (0, 1).$$

At each of the next $k - 1$ positions, the negative sum already contains $t_r = 1$, while the positive term is zero. Inductively the binary bounds force

$$t_{r+1} = \cdots = t_{r+k-1} = 0, \quad (a_i, b_i) = (0, 1) \quad (r < i < r + k). \quad (3.3)$$

At the separator $s = r + k$, the equation becomes

$$a_s = b_s + 1 - t_s. \quad (3.4)$$

We claim $t_s \leq 1$. If s cannot be the start of a valid step, then $t_s = 0$ by definition. Otherwise position $s + 1$ exists, and $t_{s+1-k} = t_{r+1} = 0$ because $k \geq 2$. The equation at $s + 1$ has a negative sum containing t_s and no positive contribution. Binary bounds therefore imply $t_s \leq 1$.

If $t_s = 0$, equation (3.4) forces $(a_s, b_s) = (1, 0)$, closing a carry block. If $t_s = 1$, it forces $a_s = b_s = c_1$, and the next $k - 1$ positions satisfy the same forced pattern as (3.3). Repeating the argument at the next separator produces a chain of starts

$$r, r + k, r + 2k, \dots, r + jk$$

with count one at each start and zero at the intervening positions. It must eventually close, because only finitely many starts are possible. The resulting two-row segment is exactly (3.1).

After its closing position, this chain makes no further contribution to (3.2): its final start is k positions before the closure. We can therefore take the next nonzero t_i , if any, and repeat the same argument. Positions between these disjoint blocks are diagonal. This proves necessity without counting normalization histories.

Sufficiency. A diagonal column changes no value. For a carry block starting at r , place $t_i = 1$ at $r, r + k, \dots, r + jk$ and zero elsewhere. The coordinate differences of the two rows are exactly the changes in (3.2). Each such formal change has weighted sum

$$w_{i+k} - (w_i + \cdots + w_{i+k-1}) = 0.$$

Thus every carry block preserves value at its actual position, and so does a concatenation of blocks and diagonal columns.

Uniqueness. Parse from the low-weight end. If the current column is diagonal, it cannot start a carry block, so it is forced to be a diagonal factor. Otherwise a block starts with k columns $(0, 1)$. At each ensuing separator, $(1, 0)$ must close the block, while an equal pair (c, c) must continue it and is followed by $k - 1$ columns $(0, 1)$. Every choice is therefore determined by the two-row word itself. $\square$

Remark 3.2. The block shapes agree with the appropriate orientation of Stanley's pair generators [1, Lemma 5.2]. The proof above is specialized to a canonical upper row and establishes exactly the version needed here. In particular, it does not assume that a numerical equality of weighted sums automatically has a unique local carry history.

4 An exact counter with only $k + 1$ coordinates

For a fixed canonical upper word, read its digits one at a time. There is a balanced state, where a completed factorization can end, and excursion states $1, \dots, k$. In an excursion, states $1, \dots, k - 1$ record progress through a required run of $(0, 1)$ columns, and state k waits at a separator. A transition selects the bottom digit as well as the next state.

Let p count paths in the balanced state, and let b_i count paths in excursion state i . The initial counter is

$$v_\varepsilon = (p, b_1, \dots, b_k) = (1, 0, \dots, 0). \quad (4.1)$$

Reading an upper zero or one acts by the following nonnegative integer linear maps:

$$Z(p, b_1, \dots, b_k) = (p, p + b_k, b_1, \dots, b_{k-1}), \quad (4.2)$$

$$O(p, b_1, \dots, b_k) = (p + b_k, b_k, 0, \dots, 0). \quad (4.3)$$

To verify the first equation, an upper zero permits either a diagonal $(0, 0)$ in the balanced state or a new excursion beginning with $(0, 1)$. An existing state $i < k$ advances to $i + 1$ on $(0, 1)$. At state k , an upper zero can only be an equal separator $(0, 0)$, returning to state 1. For an upper one, the balanced state permits the diagonal $(1, 1)$. State k either closes the excursion on $(1, 0)$ or continues it on $(1, 1)$; no other excursion state permits that upper digit. These are precisely (4.2)–(4.3).

Proposition 4.1 (The output is the coefficient). *For every canonical word $a \in \{0, 1\}^n$, the final first coordinate p obtained from (4.1)–(4.3) is $c_n(\text{val}(a))$.*

Proof. A path that ends balanced specifies a complete factorization of a pair (a, b) of the form in Theorem 3.1. That theorem identifies such pairs with all words b of the same value. Different paths cannot count the same bottom word twice, because the pair factorization is unique. Conversely every such bottom word has its factorization and therefore its path. The first coordinate counts exactly these paths. $\square$

[Kept twice: structural witness.] Proposition 4.1 produces the *exact nonnegative integer* $c_n(j)$. Theorem 9.3 below produces instead a *signed* value in $\{-1, 0, 1\}$ whose square is the parity of $c_n(j)$. Neither statement implies the other: the counter says nothing about signs, and flatness says nothing about the size of $c_n(j)$. Their consistency is asserted and checked in Section 13.2.

Example 4.2. For $k = 2$ and $n = 5$, the weights are 1, 2, 3, 5, 8. The canonical word 00001 represents 8. The counters are

$$(1, 0, 0) \xrightarrow{0} (1, 1, 0) \xrightarrow{0} (1, 1, 1) \xrightarrow{0} (1, 2, 1) \xrightarrow{0} (1, 2, 2) \xrightarrow{1} (3, 2, 0).$$

Thus $[x^8]P_5(x) = 3$. The three representations, in the same low-to-high convention, are 00001, 00110, and 11010.

The coefficient itself can be large; nevertheless its residue is computed by the same $k + 1$ coordinates reduced modulo m . No division by a residue, field structure, or primality assumption is needed.

5 Rationality for every residue, including the zero correction

5.1 The finite machine

Fix $m \geq 2$. Keep the counter v in $(\mathbb{Z}/m\mathbb{Z})^{k+1}$ and augment it by a run state $r \in \{0, 1, \dots, k\}$. The latter is the current trailing run of upper ones, capped at k . The initial state is $(v_\varepsilon, 0)$. On an upper digit, use

input	counter update	run update
0	$Z(v) \pmod{m}$	$r \mapsto 0$ provided $r < k$,
1	$O(v) \pmod{m}$	$r \mapsto \min(k, r + 1)$.

(5.1)

The zero transition from $r = k$ is rejected. Thus legal paths read exactly the canonical words. There are at most $(k + 1)m^{k+1}$ nonrejecting states; only states reachable from the initial state need be retained.

Let A be the integer adjacency matrix of this finite labeled graph: A_{uv} is the number of input labels taking state u to state v . Even though the counters live modulo m , the entries and matrix arithmetic of A live in $\mathbb{Z}$. Let λ select the initial state, and let f_a be the column vector that is one exactly when the first counter is a . Proposition 4.1 gives

$$h_{m,a}^{(k)}(n) = \lambda A^n f_a, \quad H_{m,a}^{(k)}(z) = \lambda(I - zA)^{-1} f_a. \quad (5.2)$$

Since $\det(I - zA)$ has constant term one, the inverse exists formally and has rational entries. This proves residue rationality for nonzero terms. In particular, this construction counts each exponent once, not once for each of its representations.

5.2 Internal zero coefficients

For $k \geq 3$, some exponents between zero and the degree are not represented. For instance, at $k = 3, n = 2$ the polynomial is $1 + x + x^3 + x^4$ and has a zero coefficient at exponent 2. These missing positions belong to residue zero under the full-position convention.

Put

$$Q_k(z) = 1 - z - z^2 - \dots - z^k, \quad B_k(z) = \sum_{j=2}^{k-1} (k-j)z^j, \quad (5.3)$$

where $B_2 = 0$.

Proposition 5.1 (An explicit hole generating function). *The number of zero positions in $0 \leq j \leq D_n$ has generating function*

$$J_k(z) := \sum_{n \geq 0} (D_n + 1 - s_k(n))z^n = B_k(z)S_k(z). \quad (5.4)$$

Consequently

$$\mathcal{H}_{m,a}^{(k)}(z) = H_{m,a}^{(k)}(z) + \mathbf{1}_{a=0} \frac{B_k(z)}{1 - 2z + z^{k+1}}. \quad (5.5)$$

Proof. The initial values and recurrence in (2.1) give

$$W_k(z) := \sum_{i \geq 0} w_i z^i = \frac{1 + \sum_{j=1}^{k-1} (k-j)z^j}{Q_k(z)}.$$

Since D_n is the sum of the first n weights,

$$\sum_{n \geq 0} (D_n + 1)z^n = \frac{1 + zW_k(z)}{1 - z} = \frac{1 + B_k(z)}{(1 - z)Q_k(z)} = (1 + B_k(z))S_k(z). \quad (5.6)$$

Subtract (2.5). A missing position contributes only to residue zero, which gives (5.5). $\square$

This finishes the proof of Theorem 1.1. The case $m = 1$, if included, is immediate from S_k for nonzero terms and $(1 + B_k)S_k$ for all positions. For $k = 2$ there are no holes, and the two conventions agree for every residue. At $k = 3$ the hole function is $z^2 S_3(z)$; at $k = 4$ it is $(2z^2 + z^3)S_4(z)$.

6 Exact multiplicities and a general finite-state principle

To test a coefficient against a fixed positive integer q , it is enough to distinguish $0, 1, \dots, q$ and “at least $q + 1$.” This works because all additions in the counter are nonnegative.

Set $M = q + 1$ and define $\sigma_M(t) = \min(M, t)$. For nonnegative integers,

$$\sigma_M(u + v) = \sigma_M(\sigma_M(u) + \sigma_M(v)). \quad (6.1)$$

Apply σ_M coordinatewise after every counter update. Induction using (6.1) shows that the resulting counter is the coordinatewise saturation of the exact counter, not an approximation to it. Thus its first coordinate equals q exactly when the true coefficient equals q .

There are $q+2$ possible values per coordinate, so adjoining the canonical run state gives at most $(k+1)(q+2)^{k+1}$ states. Mark first coordinate q as accepting and apply the resolvent construction (5.2). This proves Theorem 1.3. Fixed finite sets of positive multiplicities, and fixed positive thresholds, can be handled by changing the final-state indicator.

Proposition 6.1 (Finite-output principle). *Suppose a regular language contains exactly one representative of each object being counted. Suppose further that a nonnegative integer weight of a representative is read from a fixed coordinate of a finite-dimensional counter updated by fixed nonnegative integer linear maps on each input letter. Then the length generating function for objects whose weight has a fixed residue modulo m is rational. The same is true for objects with any fixed exact weight, or with weight in any fixed finite set.*

Proof. Take the product of an automaton for the regular language with the finite set of counter values modulo m . For exact weights, saturate above the largest requested weight. Saturation commutes with addition and with multiplication by a nonnegative integer, so it commutes with all the counter updates. In both cases the desired set is recognized by a finite machine and its length counts have an integer adjacency representation. $\square$

The nonnegativity hypothesis is important for saturation. If a computation can later subtract large quantities, identifying all large values would lose information. Modulo- m reduction has no analogous problem for integer linear maps, but the present exact-multiplicity conclusion does.

The significant problem-specific work is therefore not the final finite-state rationality argument. It is the construction of a regular cross-section and a correct finite-dimensional counter for the fibers. For the present weights these are supplied by Proposition 2.4 and Theorem 3.1.

7 A uniform parity automaton

The general residue construction is enough to prove rationality. To obtain a simple formula for every k , we now exploit the special structure modulo two. Let e_j be the j th unit vector in $(\mathbb{Z}/2\mathbb{Z})^k$. We describe states by $(p; b; r)$, with $b = (b_1, \dots, b_k)$ and r the canonical run state.

Scope warning. From here through Section 8, the letters $A_j, B_j, C_j, D, E_j, F, X$ denote automaton states and their generating series, not the polynomials $B_k(z)$ and $V_k(z)$, the degree D_n , or the constant C_k .

The following state families suffice:

$$\begin{aligned} A_j &= (1; e_1 + \dots + e_j; 0) & (0 \leq j \leq k), \\ B_j &= (1; e_j + \dots + e_k; 0) & (2 \leq j \leq k), \\ C_j &= (1; 0; j) & (1 \leq j \leq k), \\ D &= (0; e_1; 1), \\ E_j &= (0; e_j; 0) & (2 \leq j \leq k), \\ F &= (1; e_1; 1). \end{aligned}$$

For A_0 the excursion vector is zero. We merge $(0; e_1; 0)$ with D : these two states have first coordinate zero and identical relevant successors, namely E_2 on zero and the zero counter on one. This uses $k \geq 2$. Any zero counter, and any already forbidden input, is merged into a dead state X .

The initial state is A_0 . Every A_j, B_j, C_j , and F is accepting; D, E_j , and X are not. All transitions are specified in Table 1. There are

$$(k+1) + (k-1) + k + 1 + (k-1) + 1 + 1 = 4k + 2$$

states, including X . No minimality claim is needed.

Proposition 7.1 (Correctness of the uniform automaton). *The accepted length- n words of Table 1 are precisely the canonical words whose represented exponent has an odd coefficient in $P_n(x)$.*

Proof. The stated tuples include the initial counter and are closed under the updates (4.2)–(4.3) modulo two, with the canonical run rule. For example, if $j < k$, a zero shifts the prefix $e_1 + \dots + e_j$ right and introduces e_1 , giving A_{j+1} . From A_k , the new first excursion coordinate is $1 + 1 = 0$ modulo two, giving B_2 . Suffixes in the B family then shift until they disappear and return to A_0 .

On a one, every A_j with $j < k$ loses its excursion coordinates and goes to C_1 . From A_k or a B_j , the last excursion coordinate is one, so p changes from one to zero and the new excursion vector is e_1 , giving D . From E_k , a one changes p from zero to one and gives F . The remaining transitions follow by shifting a single e_j , or by updating the run length of a zero excursion vector. The only forbidden zero in these nondead families is the transition from C_k .

The two tuples merged into D have the same acceptance and the same successor classes, as noted above. A zero vector stays zero under both linear maps, so no continuation of it can be accepted. Hence merging it with

State or range	Input 0	Input 1
$A_j, 0 \leq j < k$	A_{j+1}	C_1
A_k	B_2	D
$B_j, 2 \leq j < k$	B_{j+1}	D
B_k	A_0	D
$C_j, 1 \leq j < k$	A_1	C_{j+1}
C_k	X	C_k
D	E_2	X
$E_j, 2 \leq j < k$	E_{j+1}	X
E_k	D	F
F	A_2	C_2
X	X	X

Table 1: The parity automaton, valid uniformly for $k \geq 2$. Empty index ranges at $k = 2$ are simply omitted; A_2, C_2, E_2, B_2 still exist.

X is legitimate even when its original run state differs. Induction on the input length proves that the table computes the same first-coordinate parity and canonicity as the general machine. Proposition 4.1 identifies that parity with the coefficient. $\square$

8 Symbolic elimination and the first proof of the odd formula

For each state in Table 1, use the same letter to denote the generating function of accepted continuations from that state. In this section, therefore, A_j denotes a series rather than a tuple. An accepting state has a constant contribution one for the empty continuation; a nonaccepting state has constant contribution zero. The dead state has series $X = 0$. Write

$$H = A_0, \quad u = z^k, \quad R_k(z) = 1 + z + \cdots + z^{k-1} = \frac{1-u}{1-z}.$$

All equations below are identities of formal power series.

8.1 The run states and the single-excursion states

The C chain satisfies $C_k = 1 + zC_k$ and $C_i = 1 + zA_1 + zC_{i+1}$ for $i < k$. Back-substitution gives

$$C_i = \frac{1}{1-z} + \frac{z(1-z^{k-i})}{1-z} A_1 \quad (1 \leq i \leq k). \quad (8.1)$$

In particular,

$$C_2 - C_1 = -z^{k-1} A_1. \quad (8.2)$$

The transitions from A_1 and F give

$$A_1 = 1 + zA_2 + zC_1, \quad F = 1 + zA_2 + zC_2.$$

Subtracting and applying (8.2),

$$F = (1-u)A_1. \quad (8.3)$$

The D, E states give $D = zE_2$, $E_i = zE_{i+1}$ for $2 \leq i < k$, and $E_k = zD + zF$. Consequently

$$D = u(D + F).$$

Since $1 - u$ is invertible as a formal power series,

$$D = uA_1. \quad (8.4)$$

8.2 Closing the two long chains

The chain beginning with A_k and continuing through $B_2, \dots, B_k$ has k accepting states, each with a one-transition to D , and then returns to H on zeros. Thus

$$A_k = R_k(1 + zD) + uH. \quad (8.5)$$

Similarly the k states $A_0, \dots, A_{k-1}$ give

$$H = R_k(1 + zC_1) + uA_k. \quad (8.6)$$

Combining (8.5) and (8.6), then canceling $1 - u$, gives

$$(1 + u)(1 - z)H = 1 + u + zC_1 + zuD. \quad (8.7)$$

The first transition equation is $H = 1 + zA_1 + zC_1$. The formula for C_1 in (8.1) then gives a second convenient identity:

$$(1 - z)H = 1 + z(1 - u)A_1. \quad (8.8)$$

Substitute $D = uA_1$ and $zC_1 = H - 1 - zA_1$ into (8.7). Its right-hand side becomes

$$u + H - z(1 - u^2)A_1.$$

Now use (8.8), multiplied by $1 + u$, to eliminate A_1 . The result is

$$(2(1 + u)(1 - z) - 1)H = 1 + 2u.$$

Replacing u by z^k proves

$$H = \frac{1 + 2z^k}{1 - 2z + 2z^k - 2z^{k+1}}.$$

Proposition 7.1 identifies H with $H_{2,1}^{(k)}$ and completes the *first* proof of Theorem 1.2.

Only factors with constant term one were canceled or inverted in this argument. In particular, the proof does not require dividing by z , and is valid at the formal-series level including the constant coefficient. The number of states is linear in k , but elimination reduces the final denominator degree to $k + 1$.

8.3 A recurrence and initial values

The rational formula yields

$$a_k(n) = 2^n \quad (0 \leq n \leq k), \quad a_k(n) = 2a_k(n-1) - 2a_k(n-k) + 2a_k(n-k-1) \quad (n \geq k+1). \quad (8.9)$$

The range $n \geq k + 1$ matters: the numerator contains the additional term $2z^k$. The first nontrivial collision occurs at length $k + 1$ and gives $a_k(k + 1) = 2^{k+1} - 2$.

9 A second, independent proof: signed cancellation

[Kept twice: Conjecture 6.3.] This section proves Theorem 1.2 again, and proves it in the greater generality of Theorem 1.4. It uses none of Sections 2–8: no canonical word, no carry block, no counter, no automaton, and no rational-function ansatz. What it buys is stated in item D1 of Section 1.5.

n	$a_2(n)$	$a_3(n)$	$a_4(n)$	$a_5(n)$	$a_6(n)$
0	1	1	1	1	1
1	2	2	2	2	2
2	4	4	4	4	4
3	6	8	8	8	8
4	8	14	16	16	16
5	12	24	30	32	32
6	20	40	56	62	64
7	32	68	104	120	126
8	48	116	192	232	248
9	72	200	356	448	488
10	112	344	660	864	960
11	176	592	1224	1668	1888
12	272	1016	2272	3220	3712

Table 2: Initial odd-coefficient counts. These values are produced by the recurrence (8.9), and independently by exact polynomial expansion, by the uniform $4k + 2$ -state automaton of Table 1, and by the signed sparse products of Section 9.

9.1 Quasifibonacci weights and coherent signings

Definition 9.1. A *quasifibonacci sequence of order k* is a sequence of positive integers $w_0, w_1, \dots$ satisfying

$$w_i > D_i := \sum_{r=0}^{i-1} w_r \quad (0 \leq i \leq k-1), \quad (9.1)$$

$$w_i = \sum_{r=i-k}^{i-1} w_r \quad (i \geq k). \quad (9.2)$$

We put $D_0 = 0$ and $D_n = \sum_{r=0}^{n-1} w_r$. The term agrees with Diao’s definition [3, Definition 1.1], with “level” replaced by “order.”

The seed condition is often called *superincreasing*: each seed is larger than all earlier seeds combined. For instance $(1, 3, 5)$ and $(2, 7, 15)$ are valid order-three seed lists; $(1, 2, 3)$ is not.

Definition 9.2. A sign sequence $\varepsilon_0, \varepsilon_1, \dots \in \{-1, 1\}$ is *coherent* if

$$\varepsilon_i = - \prod_{r=i-k}^{i-1} \varepsilon_r \quad (i \geq k). \quad (9.3)$$

Its first k signs can be chosen arbitrarily. A convenient canonical choice is

$$\varepsilon_i = \begin{cases} -1, & i \equiv k \pmod{k+1}, \\ 1, & \text{otherwise.} \end{cases} \quad (9.4)$$

Define the signed product by

$$\Phi_n(x) = \prod_{i=0}^{n-1} (1 + \varepsilon_i x^{w_i}) = \sum_{m \in \mathbb{Z}} \varphi_n(m) x^m.$$

The coefficient function $\varphi_n(m)$ vanishes outside $0 \leq m \leq D_n$. When every $\varepsilon_i = 1$ this is the ordinary product P_n of (1.2).

Theorem 9.3 (Uniform parity and signed flatness). *For every sequence in Definition 9.1, every coherent signing, and every $n \geq 0$:*

- (i) every coefficient of Φ_n is -1 , 0 , or 1 ;
- (ii) for each $m \in \mathbb{Z}$, the residue modulo two of $[x^m]P_n$ is $\varphi_n(m)^2$;
- (iii) the number of odd coefficients is independent of the seeds and of the initial signs, and satisfies

$$a_k(n) = 2^n \quad (0 \leq n \leq k), \quad (9.5)$$

$$a_k(n) = 2a_k(n-1) - 2a_k(n-k) + 2a_k(n-k-1) \quad (n \geq k+1), \quad (9.6)$$

so that $\sum_{n \geq 0} a_k(n)z^n = (1 + 2z^k)/(1 - 2z + 2z^k - 2z^{k+1})$.

The proof occupies Sections 9.2–9.6. It has two distinct parts: first flatness, then an exact norm recurrence. The latter alone would not justify counting odd coefficients without the former.

9.2 Arithmetic of the exponent sequence

Lemma 9.4 (Separation). *The weights are strictly increasing. For every $j \geq 0$,*

$$w_{j+1} > D_j, \quad 2w_j > D_j. \quad (9.7)$$

Proof. The seeds are strictly increasing by (9.1). At the first recurrence step, $w_k = D_k > w_{k-1}$. Each subsequent weight is a sum of $k \geq 2$ positive weights including its immediate predecessor, so strict increase persists.

For $0 \leq j \leq k-2$, the first inequality in (9.7) is the seed condition at index $j+1$. For $j = k-1$, we have $w_k = D_k > D_{k-1}$. If $j \geq k$, the recurrence for w_{j+1} includes the two preceding weights, so induction gives

$$w_{j+1} \geq w_j + w_{j-1} > D_{j-1} + w_{j-1} = D_j,$$

where $w_j > D_{j-1}$ is the first inequality one index lower. This proves the first inequality at all indices. For $j = 0$ the second inequality is immediate. For $j \geq 1$, combine the first inequality one index lower with strict increase:

$$D_j = D_{j-1} + w_{j-1} < w_j + w_j = 2w_j.$$

□

Lemma 9.5 (Exact overlap length). *For $n \geq k+1$, put $j = n - k - 1$. Then*

$$D_{n-1} = w_{n-1} + D_j. \quad (9.8)$$

Thus the coefficient ranges of $\Phi_{n-1}(x)$ and $x^{w_{n-1}}\Phi_{n-1}(x)$ overlap exactly in the interval $[w_{n-1}, w_{n-1} + D_j]$.

Proof. Split $D_{n-1} = w_0 + \dots + w_{n-2}$ after its first j summands. The remaining k summands are $w_{n-k-1}, \dots, w_{n-2}$, whose sum is w_{n-1} by (9.2). The interval assertion follows because $\deg \Phi_{n-1} = D_{n-1}$ and both endpoint coefficients are nonzero. Here “coefficient range” means the containing interval, not necessarily a support without internal gaps. □

Lemma 9.6 (Stanley’s seeds are admissible). *The weights $w_i = F_{i+k}^{(k)}$ of (2.1) form a quasifibonacci sequence of order k : $w_0 = 1$, $w_i = (k-1)2^{i-1} + 1$ for $1 \leq i \leq k-1$, and $w_i - D_i = k - i > 0$ for $0 \leq i \leq k-1$.*

Proof. This is (2.1) together with (2.3). Directly: the first new term after the k initial ones is k , and each subsequent initial step replaces one of those ones by the newest term, so the new value is twice its predecessor minus one. Summing $w_i = (k-1)2^{i-1} + 1$ over $1 \leq r < i$ and adding $w_0 = 1$ gives $D_i = (k-1)(2^{i-1} - 1) + i$, whence $w_i - D_i = k - i$. The later weights satisfy (9.2) by definition. □

Consequently Theorem 9.3 applied to these weights re-proves Theorem 1.2.

9.3 Coherent signs and reflection

Lemma 9.7 (Periodic description of all coherent signings). *A signing is coherent if and only if it is periodic with period dividing $k+1$ and every consecutive block of length $k+1$ has product -1 . There are exactly 2^k such signings.*

Proof. Equation (9.3) says that each length- $(k+1)$ window has product -1 . Dividing the products for two consecutive windows cancels the k common signs and gives $\varepsilon_{r+k+1} = \varepsilon_r$. Conversely, a periodic sign block with product -1 has that product in every window, hence satisfies (9.3). Its first k signs are free and its final sign is their negative product. The canonical choice (9.4) is the block of k plus signs followed by one minus sign. $\square$

Put $\Pi_n = \prod_{i=0}^{n-1} \varepsilon_i$, including $\Pi_0 = 1$.

Lemma 9.8 (Reflection). *For every integer m ,*

$$\varphi_n(D_n - m) = \Pi_n \varphi_n(m). \quad (9.9)$$

If $n \geq k+1$ and $j = n - k - 1$, then

$$\Pi_{n-1} \Pi_j = -\varepsilon_{n-1}. \quad (9.10)$$

Proof. For an individual factor, $x^{w_i}(1 + \varepsilon_i x^{-w_i}) = \varepsilon_i(1 + \varepsilon_i x^{w_i})$. Multiplication gives $x^{D_n} \Phi_n(x^{-1}) = \Pi_n \Phi_n(x)$ and proves (9.9), including the zero coefficients outside the range. Since $\Pi_j^2 = 1$, the product $\Pi_{n-1} \Pi_j$ consists precisely of the k signs of indices $j = n - k - 1$ through $n - 2$. By (9.3) its value is $-\varepsilon_{n-1}$. $\square$

9.4 The pointwise cancellation identity

This subsection contains the decisive calculation. It is an identity over the integers, before reducing anything modulo two.

Proposition 9.9 (Two disjoint tails). *Let $n \geq k+1$, $j = n - k - 1$, and $v = w_j$. For $0 \leq u \leq D_j$,*

$$\boxed{\varphi_n(w_{n-1} + u) = \varepsilon_{n-1} \varepsilon_j (\varphi_j(u - v) - \varphi_j(u + v))}. \quad (9.11)$$

The two terms on the right cannot both be nonzero.

Proof. By Lemma 9.4, $w_{j+1} > D_j$, so the factors of index $j+1$ or higher have all their nonconstant terms in degrees larger than D_j . Therefore, on the interval $0 \leq u \leq D_j$,

$$\varphi_{n-1}(u) = \varphi_j(u) + \varepsilon_j \varphi_j(u - v). \quad (9.12)$$

There is no convergence or truncation approximation in this equation: later factors simply cannot contribute to such a low exponent.

Reflect the high end of Φ_{n-1} using $D_{n-1} = w_{n-1} + D_j$ from Lemma 9.5 and (9.12). We obtain

$$\begin{aligned} \varphi_{n-1}(w_{n-1} + u) &= \Pi_{n-1} \varphi_{n-1}(D_{n-1} - w_{n-1} - u) \\ &= \Pi_{n-1} \varphi_{n-1}(D_j - u) \\ &= \Pi_{n-1} (\varphi_j(D_j - u) + \varepsilon_j \varphi_j(D_j - u - v)) \\ &= \Pi_{n-1} \Pi_j (\varphi_j(u) + \varepsilon_j \varphi_j(u + v)) \\ &= -\varepsilon_{n-1} (\varphi_j(u) + \varepsilon_j \varphi_j(u + v)). \end{aligned} \quad (9.13)$$

Now use $\Phi_n = \Phi_{n-1} + \varepsilon_{n-1} x^{w_{n-1}} \Phi_{n-1}$ and add (9.13) to ε_{n-1} times (9.12). The two copies of $\varphi_j(u)$ cancel exactly, leaving (9.11).

For the first tail to be nonzero, $u \geq v$ is necessary. For the second, $u \leq D_j - v$ is necessary. These conditions cannot hold simultaneously, because $2v = 2w_j > D_j$ by Lemma 9.4. More explicitly, the possible tail intervals are

$$[0, D_j - v] \quad \text{and} \quad [v, D_j],$$

with an interval understood to be empty if its endpoints are reversed. They are disjoint. $\square$

Remark 9.10. At the very first overlap, $n = k + 1$ and $j = 0$. Both tails are empty: $\varphi_0(-w_0) = \varphi_0(w_0) = 0$. Thus the only overlapping coefficient, at exponent $w_k = D_k$, cancels. This is an important endpoint check on the indexing.

9.5 Flatness and the parity lift

Proof of Theorem 9.3(i)–(ii). For $0 \leq n \leq k$, the seed condition ensures that two distinct subsets of $\{w_0, \dots, w_{n-1}\}$ cannot have the same sum. Indeed, at their largest index of disagreement, the larger seed exceeds the sum of all earlier seeds. Every nonzero coefficient of Φ_n is therefore a single product of signs and is 1 or -1 .

Proceed by strong induction on n . For $n \geq k + 1$, outside $[w_{n-1}, D_{n-1}]$ at most one of the two terms Φ_{n-1} and $\varepsilon_{n-1}x^{w_{n-1}}\Phi_{n-1}$ contributes. The coefficient there is zero or, up to sign, an earlier coefficient. Inside the overlap, Proposition 9.9 expresses it as the signed difference of two coefficients of Φ_j , where $j < n$, with at most one nonzero term. The induction hypothesis consequently proves flatness everywhere.

Modulo two, $\varepsilon_i \equiv 1$, so $P_n \equiv \Phi_n$ in $\mathbb{F}_2[x]$. A flat integer coefficient is odd exactly when its square is one. Therefore

$$([x^m]P_n \bmod 2) = \varphi_n(m)^2 \in \{0, 1\}. \quad (9.14)$$

This proves both assertions. $\square$

The result can also be read as a signed enumeration statement. Among all subsets having a specified weight, split them according to the product of their assigned signs. The sizes of the two classes differ by at most one. This signing is not necessarily the parity of the subset's cardinality. For odd k , that distinction is exactly what makes the theorem possible.

Corollary 9.11 (An infinite-product extension). *For any admissible seeds and coherent signing, the formal infinite product $\prod_{i \geq 0} (1 + \varepsilon_i x^{w_i})$ has all coefficients in $\{-1, 0, 1\}$.*

Proof. The weights tend to infinity. For each fixed exponent, all sufficiently late factors contribute only their constant term, so the coefficient stabilizes to one of the finite-product coefficients already proved flat. $\square$

For even k , choosing all initial signs negative makes every sign negative; this recovers the setting of Diaó's infinite-product theorem [3]. For odd k , a periodic coherent signing replaces that all-minus choice.

9.6 The exact norm recurrence

Define

$$q_n = \|\Phi_n\|_2^2 := \sum_{m \in \mathbb{Z}} \varphi_n(m)^2, \quad G_n = \sum_{m \in \mathbb{Z}} \varphi_{n-1}(w_{n-1} + m) \varphi_{n-1}(m).$$

All sums here and below are finite. Expanding the square in $\Phi_n = \Phi_{n-1} + \varepsilon_{n-1}x^{w_{n-1}}\Phi_{n-1}$ gives

$$q_n = 2q_{n-1} + 2\varepsilon_{n-1}G_n. \quad (9.15)$$

Flatness and (9.14) imply $q_n = a_k(n)$, but we first derive the norm recurrence directly.

Lemma 9.12 (The overlap inner product). *For $n \geq k + 1$ and $j = n - k - 1$,*

$$\varepsilon_{n-1}G_n = q_j - q_{j+1}. \quad (9.16)$$

Proof. Only $0 \leq u \leq D_j$ can contribute to G_n . Substitute (9.12) and (9.13), abbreviating $v = w_j$ and $e = \varepsilon_j$:

$$\varepsilon_{n-1}G_n = - \sum_{u=0}^{D_j} (\varphi_j(u) + e \varphi_j(u+v)) (\varphi_j(u) + e \varphi_j(u-v)).$$

The product $\varphi_j(u+v)\varphi_j(u-v)$ vanishes for every u , by the same separation $2v > D_j$ used in Proposition 9.9. The two mixed sums are equal after a change of index. If

$$T = \sum_{u \in \mathbb{Z}} \varphi_j(u)\varphi_j(u-v),$$

then the preceding display becomes $\varepsilon_{n-1}G_n = -q_j - 2eT$. On the other hand, multiplication of Φ_j by $1 + ex^v$ yields $q_{j+1} = 2q_j + 2eT$. Eliminating T proves $\varepsilon_{n-1}G_n = q_j - q_{j+1}$. $\square$

Proof of Theorem 9.3(iii). The superincreasing seed block gives $q_n = 2^n$ for $0 \leq n \leq k$. Substituting (9.16) into (9.15) gives

$$q_n = 2q_{n-1} - 2q_{n-k} + 2q_{n-k-1} \quad (n \geq k+1).$$

Since $q_n = a_k(n)$, this proves (9.5)–(9.6). Neither the coefficients of this recurrence nor its initial values depend on the seeds or on the first k signs.

Finally multiply $\sum_n a_k(n)z^n$ by $V_k(z) = 1 - 2z + 2z^k - 2z^{k+1}$. Its constant coefficient becomes one, its coefficients of degrees $1, \dots, k-1$ vanish, its coefficient of degree k is two, and all later coefficients vanish by the recurrence. Thus the product is $1 + 2z^k$. $\square$

By Lemma 9.6, this is a second and independent proof of Theorem 1.2, and it also proves the seed-independent Theorem 1.4. The two proofs share no lemma: the finite-state route never forms a signed product, and the signed route never forms a canonical word.

9.7 Examples, and the role of the signs

The first genuinely overlapping order-three example. The standard order-three weights are $1, 3, 5, 9, 17, 31, 57, 105, 193, \dots$. The canonical coherent signs repeat $+, +, +, -$. For the first four factors,

$$\begin{aligned} \Phi_4(x) &= (1+x)(1+x^3)(1+x^5)(1-x^9) \\ &= 1 + x + x^3 + x^4 + x^5 + x^6 + x^8 \\ &\quad - x^{10} - x^{12} - x^{13} - x^{14} - x^{15} - x^{17} - x^{18}. \end{aligned}$$

There are fourteen nonzero coefficients. The ordinary all-plus product has a coefficient two at x^9 and fourteen odd coefficients elsewhere. The recurrence agrees: $a_3(4) = 2 \cdot 8 - 2 \cdot 2 + 2 \cdot 1 = 14$, matching row $n = 4$, column $k = 3$ of Table 2.

A later example displays the disjoint-tail mechanism rather than just a single cancellation. At $n = 8$ we have $j = 4$, $v = w_4 = 17$ and $D_4 = 18$. The overlap is $105 \leq m \leq 123$. Formula (9.11) allows nonzero coefficients there only at $m = 105, 106, 122, 123$; each is -1 . The fifteen intermediate positions cancel.

Why the naive all-minus choice fails. For the same weights,

$$[x^9](1-x)(1-x^3)(1-x^5)(1-x^9) = -2.$$

The contributions come from the singleton $\{9\}$ and the triple $\{1, 3, 5\}$; both have negative cardinality sign. An odd-order recurrence replaces k selected parts by one part without changing cardinality parity. The coherent signing instead forces those two contributions to have opposite signs. This explains why the proof must change the signing, not merely reuse the all-minus Fibonacci product.

An OEIS connection. For $k = 2$ the sequence satisfies $a_2(n) = 2 \text{ A104767}(n)$ for $n \geq 1$, with the separate value $a_2(0) = 1$. The cited OEIS entry records the corresponding Fibonacci signed-product interpretation [5]. No new OEIS identifier is assigned or implied for the tables accompanying this manuscript.

Remark 9.13 (The seed hypothesis cannot simply be omitted). **[Kept twice: initialization boundaries.]** For $k = 3$ and seeds $(1, 2, 3)$, the first three all-plus factors already have an even coefficient at exponent three and only six odd coefficients, not 2^3 . Likewise, duplicate order-two seeds $(1, 1)$ give $(1 + x)^2$, with two odd coefficients instead of four. Theorem 1.4 is not asserting the same initial values for arbitrary positive seed lists. This is a different degeneration from the $k = 1$ failure discussed in Section 14.2, which concerns the order rather than the seeds.

10 A positive recurrence and a finite counting formula

The minimal recurrence (8.9) has a negative coefficient. A slightly longer one has only nonnegative coefficients and makes both positivity and the analytic asymptotics of Section 11.1 more transparent. Put

$$R_k(z) = 1 + z + \cdots + z^{k-1}, \quad L_k(z) = z + z^2 + \cdots + z^{k-1} + 2z^{2k}, \quad N_k(z) = 1 + 2z^k.$$

Direct multiplication gives

$$R_k(z)V_k(z) = 1 - L_k(z). \quad (10.1)$$

Indeed $(1 - z)R_k = 1 - z^k$ and $V_k = (1 - 2z) + 2z^k(1 - z)$, so $R_k V_k = R_k - 2zR_k + 2z^k(1 - z^k) = 1 - z - \cdots - z^{k-1} - 2z^{2k}$.

Corollary 10.1 (Positive recurrence). *For every $n \geq 2k$,*

$$a_k(n) = \sum_{r=1}^{k-1} a_k(n-r) + 2a_k(n-2k). \quad (10.2)$$

Equivalently,

$$\sum_{n \geq 0} a_k(n)z^n = \frac{N_k(z)R_k(z)}{1 - L_k(z)}. \quad (10.3)$$

Proof. Multiply numerator and denominator of (1.7) by R_k and use (10.1). The numerator has degree $2k - 1$, so extracting coefficients in degrees at least $2k$ proves (10.2). $\square$

Formula (10.3) is the generating function of the following simple objects. Choose a prefix of size $d + ek$, where $0 \leq d < k$ and $e \in \{0, 1\}$; the prefix has one color if $e = 0$ and two colors if $e = 1$. Follow it by an ordered sequence of parts of sizes $1, 2, \dots, k-1, 2k$, where the part $2k$ has two colors and the other parts each have one color. Total size is counted by z .

This is an exact equinumerosity obtained from the generating function. It is *not* claimed to be an explicit bijection with the odd-coefficient positions of the original product.

In particular, an entirely finite, nonrecursive expression is

$$a_k(n) = \sum_{e=0}^1 \sum_{d=0}^{k-1} \sum_{\substack{m_1, \dots, m_{k-1}, b \geq 0 \\ \sum_{r=1}^{k-1} rm_r + 2kb = n - d - ek}} 2^{e+b} \frac{(m_1 + \cdots + m_{k-1} + b)!}{m_1! \cdots m_{k-1}! b!}. \quad (10.4)$$

An impossible constraint contributes zero. The multinomial coefficient orders the chosen parts; the power of two supplies their colors and, when applicable, the prefix color. This proves (10.4) directly from the description above.

Proposition 10.2 (Minimal recurrence order). *Over $\mathbb{Q}$, the smallest order of an eventual homogeneous constant-coefficient recurrence for $a_k(n)$ is $k + 1$.*

Proof. The denominator V_k has degree $k + 1$. It is relatively prime to $N_k(z) = 1 + 2z^k$, because

$$V_k(z) = (1 - 2z)N_k(z) + 2z^{k+1}.$$

A common root would have to be zero, but $N_k(0) = 1$. Thus the fraction in (1.7) is reduced. If a recurrence with denominator polynomial $Q(z)$ holds eventually, then $Q(z) \sum a_k(n)z^n$ is a polynomial, so the reduced denominator V_k divides Q . Its degree is therefore a lower bound for the recurrence order, and the recurrence (8.9) attains it. $\square$

11 Asymptotics: growth twice, and two sparsity statements

Let β_k be the positive root greater than one of

$$\beta_k^k = \beta_k^{k-1} + \cdots + \beta_k + 1, \quad (11.1)$$

equivalently $t_k = 1/\beta_k$ is the root in $(0, 1)$ of $Q_k(z) = 1 - z - \cdots - z^k$; and let r_k be the root in $(0, 1)$ of $V_k(z) = 1 - 2z + 2z^k - 2z^{k+1}$, with $\rho_k = 1/r_k$. By (10.1) and $R_k > 0$ on $(0, 1)$, the number r_k is simultaneously the unique positive solution of $L_k(r) = 1$; the two routes below therefore concern the same constant.

11.1 The analytic route to the dominant root

[Kept twice: dominant-root asymptotic, proof 1 of 2.]

Theorem 11.1 (Asymptotic count, analytic proof). *For every fixed $k \geq 2$ there is $0 < \theta_k < \rho_k$ with*

$$a_k(n) = C_k \rho_k^n + O(\theta_k^n), \quad C_k = \frac{N_k(r_k)R_k(r_k)}{r_k L'_k(r_k)} > 0. \quad (11.2)$$

Moreover $1 < \rho_k < 2$, and ρ_k is the unique real root in $(1, 2)$ of

$$\rho^{k+1} - 2\rho^k + 2\rho - 2 = 0. \quad (11.3)$$

Proof. L_k is strictly increasing on the positive real axis, with $L_k(0) = 0$ and $L_k(1) = k + 1 > 1$. Thus $r_k \in (0, 1)$ exists uniquely and $L'_k(r_k) > 0$. Also

$$L_k(1/2) = 1 - 2^{1-k} + 2^{1-2k} < 1,$$

so $r_k > 1/2$ and $\rho_k < 2$.

For complex z with $|z| < r_k$ the triangle inequality gives $|L_k(z)| \leq L_k(|z|) < 1$. If $|z| = r_k$ and $L_k(z) = 1$, equality in the triangle inequality forces all summands to have the same argument, and their sum is positive real. The term z occurs with positive coefficient, so z itself is positive real, whence $z = r_k$. Therefore r_k is the only zero of $1 - L_k$ on its smallest root circle, and it is simple.

By (10.3) the singular part of the generating function at r_k is $C_k/(1 - z/r_k)$ with C_k as displayed. Subtracting it leaves a rational function analytic in a disk of radius strictly larger than r_k ; Cauchy's coefficient bound on a circle inside that disk gives the exponentially smaller error term. Finally, (10.1) and $R_k(r_k) > 0$ give $V_k(r_k) = 0$; substituting $r_k = 1/\rho_k$ yields (11.3), and uniqueness follows from the uniqueness of the positive root of $1 - L_k$ together with positivity of R_k on the positive axis. $\square$

11.2 The combinatorial route to the dominant root, and strict sparsity among nonzero coefficients

[Kept twice: dominant-root asymptotic, proof 2 of 2; and sparsity statement 1 of 2.]

Theorem 11.2 (Growth and strict sparsity among nonzero coefficients). *For fixed $k \geq 2$,*

$$1 < \rho_k < \beta_k < 2, \quad a_k(n) = \gamma_k \rho_k^n + O(\theta_k^n) \quad (11.4)$$

for some $0 < \theta_k < \rho_k$, where

$$\gamma_k = -\frac{1 + 2r_k^k}{r_k V'_k(r_k)} > 0. \quad (11.5)$$

In particular $a_k(n)/s_k(n)$ tends to zero exponentially, where $s_k(n)$ is the number of nonzero coefficients of P_n .

Proof. First, $V_k(0) = 1$ and $V_k(1) = -1$. On $[0, 1]$,

$$V'_k(z) = -2 + 2z^{k-1}(k - (k+1)z) < 0,$$

because the maximum of $z^{k-1}(k - (k+1)z)$ is $((k-1)/(k+1))^{k-1} < 1$. Thus r_k exists uniquely and is simple.

To exclude other poles of the same modulus, use the automaton of Table 1 rather than complex analysis. Remove X and C_k from its graph. The remaining states form one strongly connected component: every state is reachable from A_0 , and the transition table supplies a return path to A_0 . There is an all-zero cycle of length $2k$ through the A and B families. Replacing its first edge $A_0 \rightarrow A_1$ with $A_0 \rightarrow C_1 \rightarrow A_1$ gives a cycle of length $2k+1$. The component is therefore aperiodic. Its spectral radius is strictly greater than one: at a common multiple of those two lengths there are two different closed walks, whose arbitrary concatenations already give exponential growth.

The Perron–Frobenius theorem for a finite irreducible aperiodic nonnegative matrix now gives a simple strictly dominant positive eigenvalue. The other productive component, the loop at C_k , has spectral radius one. Both the initial state and accepting states meet the main component, so its leading contribution is nonzero. Hence the rational series has one simple dominant pole, necessarily r_k , and coefficient extraction gives (11.4)–(11.5). Subdominant polynomial factors are absorbed by choosing θ_k strictly above the moduli of all other contributing eigenvalues.

Let $t_k = 1/\beta_k$, the unique root in $(0, 1)$ of Q_k . Its simplicity and the triangle inequality show that no other root of Q_k has modulus t_k : equality would force the terms z and z^2 to have the same positive argument, hence $z = t_k$. Equation (2.5) therefore gives $s_k(n) \sim \delta_k \beta_k^n$ with $\delta_k > 0$. Also $1/2 < t_k \leq 1/\beta_2 < 2/3$ and $1 - 2t_k + t_k^{k+1} = 0$, so

$$V_k(t_k) = t_k^k(2 - 3t_k) > 0.$$

Since V_k is strictly decreasing, $r_k > t_k$, that is $\rho_k < \beta_k$. The remaining inequalities are immediate from the locations of r_k, t_k . Taking the ratio of the two asymptotic formulas proves exponential sparsity among nonzero coefficients. $\square$

Corollary 11.3 (The two leading constants agree). $C_k = \gamma_k$.

Proof. Differentiate (10.1): $R'_k V_k + R_k V'_k = -L'_k$. At $z = r_k$ we have $V_k(r_k) = 0$, so $R_k(r_k) V'_k(r_k) = -L'_k(r_k)$. Substituting into (11.2) gives

$$C_k = \frac{N_k(r_k) R_k(r_k)}{r_k L'_k(r_k)} = \frac{N_k(r_k) R_k(r_k)}{-r_k R_k(r_k) V'_k(r_k)} = -\frac{N_k(r_k)}{r_k V'_k(r_k)} = \gamma_k. \quad \square$$

Both expressions are retained in Tables 3 and 4, each computed from its own defining formula. Agreement of the two columns is a further arithmetic check on the merged data.

[Kept twice: the inequality $\rho_k < \beta_k$.] Theorem 11.2 obtained it from $V_k(t_k) = t_k^k(2 - 3t_k) > 0$; Section 11.3 obtains it again from $L_k(1/\beta_k) < 1$, without reference to V_k .

11.3 Zero density in the full coefficient range

[Kept twice: sparsity statement 2 of 2.] The following theorem is *not* Theorem 11.2 restated. Its denominator counts every position from exponent zero to the degree, including positions whose coefficient is zero. The signed route has no access to the support generating function (2.5) and therefore cannot prove Theorem 11.2; conversely the statement below holds for arbitrary superincreasing seeds, where $s_k(n)$ is not given by (2.5) at all.

Theorem 11.4 (Density among all positions). *Let $w_0, w_1, \dots$ be any quasifibonacci sequence of order k . Then $D_n = \Theta(\beta_k^n)$, $\rho_k < \beta_k$, and*

$$\frac{a_k(n)}{D_n + 1} = O((\rho_k/\beta_k)^n) \longrightarrow 0. \quad (11.6)$$

Proof. The number $\beta_k \in (1, 2)$ satisfies $\sum_{i=1}^k \beta_k^{-i} = 1$, the growth equation of the exponent sequence. For every admissible seed list, $D_n = \Theta(\beta_k^n)$. One sees this without a spectral theorem: choose positive constants bounding w_i/β_k^i above and below for the first k indices, and propagate the bounds through (9.2). Summing preserves the same exponential order.

Put $s = \beta_k^{-1} = t_k$. Since the k terms $s, \dots, s^k$ sum to one, we have $s^k < 1/2$. Consequently

$$L_k(s) = (s + \dots + s^{k-1}) + 2s^{2k} = 1 - s^k + 2s^{2k} < 1,$$

so $s < r_k$ and $\rho_k < \beta_k$. Combining with Theorem 11.1 gives (11.6). $\square$

The denominator of (11.6) counts every coefficient position from zero to the degree, including positions with coefficient zero. *No assertion about density among nonzero coefficients only is implicit in (11.6)*; that separate statement is Theorem 11.2, and it is proved only for Stanley's initialization, where the support count (2.5) is available.

11.4 Large-order expansion

There is also a useful asymptotic as $k \rightarrow \infty$, with no counterpart on the finite-state route:

$$\rho_k = 2 - 2^{1-k} - (2k - 4)4^{-k} + O(k^2 8^{-k}). \quad (11.7)$$

Here is a justification that keeps the varying parameter under control. Write $t = 2^{-k}$ and $\delta = 2 - \rho_k$. Equation (11.3) becomes

$$\delta = 2t(1 - \delta)(1 - \delta/2)^{-k}. \quad (11.8)$$

For all sufficiently large k , the equivalent function $f(\delta) = \delta(2 - \delta)^k - 2 + 2\delta$ is negative at zero and positive at $4t$, since $f(4t) = 4(1 - 2t)^k - 2 + 8t \rightarrow 2$. The uniqueness already proved places the relevant root in $(0, 4t)$. Thus $k\delta \rightarrow 0$, and Taylor expansion, uniformly on that interval, gives

$$(1 - \delta)(1 - \delta/2)^{-k} = 1 + \frac{k-2}{2}\delta + O(k^2 \delta^2).$$

First (11.8) yields $\delta = 2t + O(kt^2)$; substituting back gives $\delta = 2t + 2(k-2)t^2 + O(k^2 t^3)$, proving (11.7). This expansion is separate from the fixed- k , large- n asymptotic of Theorem 11.1.

11.5 Two numerical tables

[Kept twice: numerical growth constants.] Table 3 is the primary record, computed by bisection at 80-digit working precision with Python's `Decimal` arithmetic, using the formulas of Theorem 11.1. Table 4 is an independent recomputation in double-precision floating point, using instead the formulas of Theorem 11.2. The two agree to floating-point precision; for instance $\rho_2 = 1.54368901269207636157\dots$ against 1.5436890126920766.

Both tables are illustrative numerical approximations, *not* certified interval enclosures. All the strict inequalities and asymptotic statements above are proved without numerical root finding.

12 Exact algorithms

[Kept twice: odd-count algorithm families, and coefficient oracles.] Two families of algorithms are supplied, descending from the two proofs. Section 13.2 explains why their agreement is informative while the agreement of two members of the same family is not.

k	ρ_k	C_k	β_k
2	1.543689012692	1.473679689278	1.618033988750
3	1.716672749282	1.550023534317	1.839286755214
4	1.855679027977	1.362757096516	1.927561975483
5	1.930604988558	1.200842855191	1.965948236645
6	1.966580689288	1.109329020459	1.983582843424
7	1.983724513901	1.059879860829	1.991964196605
8	1.991997413559	1.032931667570	1.996031179735

Table 3: Primary numerical record: 80-digit **Decimal** bisection, truncated for display. Here ρ_k is the odd-coefficient growth constant, C_k the leading constant of (11.2), and β_k the k -bonacci growth constant.

k	Support growth β_k	Odd growth ρ_k	γ_k
2	1.618033989	1.543689013	1.473679689
3	1.839286755	1.716672749	1.550023534
4	1.927561975	1.855679028	1.362757097
5	1.965948237	1.930604989	1.200842855
6	1.983582843	1.966580689	1.109329020
8	1.996031180	1.991997414	1.032931668
10	1.999018633	1.998031429	1.009930808

Table 4: Independent recomputation by floating-point bisection from (11.5). Compare γ_k here with C_k in Table 3; Corollary 11.3 proves they are equal.

12.1 Counting without expanding a polynomial

The recurrence (8.9) computes $a_k(0), \dots, a_k(N)$ in $O(N)$ integer arithmetic operations. A streaming implementation needs only $k + 1$ past values; the supplied `count_prefix` returns the whole list and so intentionally uses $O(N)$ integer storage.

For an isolated very large index, let

$$\chi_k(X) = X^{k+1} - 2X^k + 2X - 2.$$

Compute $X^N \bmod \chi_k(X)$ by binary powering. If the remainder is $\sum_{i=0}^k b_i X^i$, then

$$a_k(N) = \sum_{i=0}^k b_i 2^i. \quad (12.1)$$

To justify this, let E be the shift operator on the sequence. The proved recurrence says $\chi_k(E)a = 0$ beginning at index zero in shift-operator form, so polynomial division applied to E^N gives (12.1). Naive polynomial multiplication and reduction use $O(k^2 \log(N + 1))$ ring operations. The same code works modulo any positive integer; it does not assume the modulus is prime.

These are arithmetic-operation bounds, not bit-complexity bounds. In exact arithmetic $a_k(N)$ has $\Theta(N)$ bits for fixed k , by Theorem 11.1; writing it already takes linear time in its bit length.

12.2 A single coefficient's parity in linearly many steps

Proposition 9.9 gives an efficient coefficient oracle. Precompute the weights, partial sums, and signs through index $n - 1$. To evaluate $\varphi_n(m)$, first return zero if $m \notin [0, D_n]$. For $n > k$:

- (1) if $m < w_{n-1}$, replace (n, m) by $(n - 1, m)$;
- (2) if $m > D_{n-1}$, replace (n, m) by $(n - 1, m - w_{n-1})$ and multiply the accumulated sign by ε_{n-1} ;
- (3) otherwise set $j = n - k - 1$, $u = m - w_{n-1}$, $v = w_j$. Formula (9.11) gives at most one continuing branch: use $\varphi_j(u - v)$ if $u \geq v$; use $-\varphi_j(u + v)$ if $u + v \leq D_j$; otherwise return zero. In either continuing case multiply also by $\varepsilon_{n-1}\varepsilon_j$.

When the index is at most k , a greedy subset-sum test on the superincreasing seeds finishes the computation. At every step the index strictly decreases; there is no branching tree. Hence the query uses $O(n)$ arithmetic comparisons, additions, and sign changes after $O(n + k)$ -entry preprocessing, including validation of the seed block. The absolute value of the resulting signed coefficient is the requested parity of the all-plus coefficient.

For fixed k and seed bit length b , the stored exponents have $O(n + b)$ bits. The arithmetic-step description therefore does not assert constant-time operations on arbitrarily large exponents. Its principal advantage is avoiding storage proportional to the exponentially large degree.

12.3 The exact integer coefficient from the counter

The counter of Section 4 gives a different oracle with a different output: for Stanley's seeds it returns the exact nonnegative integer $c_n(j)$, not a sign. Given any binary word, normalize it and run (4.1)–(4.3). The supplied function is `coefficient_for_canonical`; for example `coefficient_for_canonical((0,0,0,0,1), k=2)` returns 3, in agreement with the worked example of Section 4. Neither oracle subsumes the other: the signed oracle accepts arbitrary superincreasing seeds and arbitrary initial sign blocks but returns only a value in $\{-1, 0, 1\}$; the counter oracle returns the exact multiplicity but is tied to (2.1).

12.4 Direct multiplication as an independent check

An integer bitset represents a polynomial over $\mathbb{F}_2$ by storing the coefficient of x^m in bit m . Multiplication by $1 + x^w$ is exactly

```
bits ^= bits << w
```

and the final odd-coefficient count is `bits.bit_count()`. This is an excellent small-instance oracle, but its memory requirement is proportional to the polynomial degree; the supplied implementation has an explicit degree cap to prevent accidental enormous allocations.

The signed sparse and dense checks perform integer polynomial multiplication, not just multiplication modulo two. They therefore test the stronger flatness claim and the sign-sensitive overlap identity as well. The reference expansion `expand` of the finite-state package is a third and separate implementation: it begins with the coefficient vector `[1]` and multiplies by each factor $1 + x^{w_i}$ using integer additions at offsets zero and w_i , using neither normalization, nor the counter, nor a guessed recurrence.

12.5 Complexity of the finite-state machines

Once a machine with N states has been constructed, advancing its count vector by one length uses at most $2N$ integer additions, because every state has at most two labeled successors. Thus the first L coefficient histograms can be generated in $O(LN)$ integer-arithmetic operations after graph construction. This is again an arithmetic-operation bound, not a bit-complexity bound: the counts themselves grow with L . Constructing each successor updates $k + 1$ counter coordinates, so a straightforward breadth-first construction uses $O(kN)$ such coordinate operations plus dictionary overhead. Very large parameters can still produce a large machine; the supplied code raises an error when its explicit state limit is exceeded rather than returning a truncated graph.

13 Exact computation, cross-verification, and finite certificates

The general proofs above do not require a computer. Computation serves three different purposes here: detecting implementation or indexing errors; making the constructions reusable; and certifying a specified rational expression once a finite linear representation is known. The third purpose is stronger than checking a long list of initial terms.

13.1 Two verification suites

[Kept twice: verification suites and their coverage tables.] The two suites test disjoint objects and are reported separately, because a single merged summary would destroy the record of what was actually checked.

Check	Parameter range	Comparisons
Residue histograms	$2 \leq k \leq 8, 0 \leq n \leq 16, m = 2, 3, 4, 5$	476
Individual residue entries	All residues in those histograms	1,666
Binary normalizations and pair factorizations	$2 \leq k \leq 8$, all words of lengths 0 through 11	28,665
Complete representation fibers	Same exhaustive word ranges	22,582
Exact multiplicities	$2 \leq k \leq 5, 1 \leq q \leq 4, 0 \leq n \leq 12$	208
Parity sequence entries	$2 \leq k \leq 25, 0 \leq n \leq 250$	6,024
Parity quotient transitions	All reachable generic modulo-two states, $2 \leq k \leq 25$	3,384

Table 5: Coverage of `code/verify.py`, the suite attached to the finite-state proof. Recorded in `data/verification_summary.json`.

The largest directly expanded polynomial had degree 227,138. Every residue comparison includes the zero-position correction (5.5). The exhaustive pair test parses each carry block independently and checks that it preserves value at its actual weight positions. The parity test compares the uniform table both with the generic modulo-two machine and with recurrence (8.9).

Check	Recorded coverage
Direct bitset products	519 prefix comparisons for $2 \leq k \leq 25$, with an 8,000,000-degree cap; the maximum n depends on k .
Signed dense products	144 products: $2 \leq k \leq 10$, $1 \leq n \leq 16$; 19,338 coefficient-oracle comparisons.
Every initial signing	All 2^k choices for $2 \leq k \leq 6$, two seed families, $n = 12$: 248 cases.
Random seed/sign cases	96 superincreasing seed lists and initial signings, each at $n = 12$, with a fixed random seed.
Pointwise overlap formula	5,821 exact coefficient identities for $2 \leq k \leq 6$, through $n = 14$.
Positive recurrence	28,101 comparisons for $2 \leq k \leq 30$, through $n = 1000$.
Fast versus linear counts	Eight indices per k for $2 \leq k \leq 30$, in exact arithmetic and modulo 1,000,000,007.

Table 6: Coverage of `code/verify_signed.py`, the suite attached to the signed-cancellation proof. Recorded in `data/verification_signed.json`.

Coverage is intentionally reported with its limitations. The degree cap means that the largest tested orders in the bitset sweep reach only the nonoverlapping initial block; they are *not* evidence of long-run behavior at those orders. Nontrivial overlap checks occur at the smaller orders, and the proofs, not the sweep, establish all orders and all indices. Equally, agreement between the two recurrence-based counting routines (`count_prefix` and `count_fast`) checks their implementation, not the combinatorial theorem: both descend from (8.9).

The generated data include odd counts through $n = 200$ for $2 \leq k \leq 20$, 80-digit working-precision growth constants, complete decimal values of $a_k(10000)$ at six orders, and large-index modular values. Some compact checkpoints are shown below.

k	Decimal digits of $a_k(10000)$	$a_k(10^{12}) \bmod 1000000007$
2	1886	338509426
3	2348	699562826
4	2686	896800000
5	2858	178164254
10	3007	587625022
20	3011	367480980

Table 7: Large-index checkpoints. Each value was obtained both by the linear recurrence (8.9) and by the polynomial powering (12.1). The complete integers are in `data/large_values.json`. Because both algorithms descend from (8.9), their agreement checks implementation, not the combinatorial theorem.

The two original packages recorded their runs under Python 3.13.5; the merged archive’s regenerated JSON records were produced under Python 3.14.4, and, for four optional symbolic elimination identities, SymPy

1.14.0. Python 3.10 or newer suffices. The main tests and certificates need only the Python standard library. The finite-state suite uses explicit error checks rather than removable Python assertions, so running Python with optimization enabled does not disable validation; the signed suite still uses `assert` and refuses to run under `python -O`.

13.2 The merged cross-verification

[Kept twice: count tables, and the two coefficient oracles.] The script `code/cross_check.py` exists only in the merged archive: neither original package could run these comparisons, because each held only one of the two implementations. It records the following.

(X1) *Structural witnesses agree.* For $2 \leq k \leq 6$ and $0 \leq n \leq 12$, for every exponent $0 \leq j \leq D_n$,

$$|\varphi_n(j)| = c_n(j) \bmod 2,$$

with $\varphi_n(j)$ from `CoefficientOracle.signed` and $c_n(j)$ from the exact dense expansion; and for every represented j , `coefficient_for_canonical` applied to the normalized word reproduces $c_n(j)$. This links item D2 of Section 1.5: flatness and the fiber counter are different facts about the same coefficient.

(X2) *The two proofs agree numerically.* For $2 \leq k \leq 8$ and $0 \leq n \leq 100$, the polynomial-powering value (12.1) equals the value produced by running the uniform $4k + 2$ -state automaton of Table 1. These two routines descend from different proofs, so this is an independent check, unlike `count_prefix` versus `count_fast`.

(X3) *The two count tables agree.* `data/counts.csv` and `data/odd_counts.csv` were generated by the two packages independently; every one of their 707 common (k, n) entries is compared.

(X4) *The two growth tables agree.* The 80-digit `Decimal` values in `data/growth.csv` and the floating-point values in `data/growth_constants.csv` agree to floating-point precision, and C_k matches γ_k as Corollary 11.3 requires.

(X5) *The weight conventions agree.* The two modules' weight functions produce the same sets for $2 \leq k \leq 12$, confirming the index-shift reconciliation of Section 1.6.

The outcome is stored in `data/cross_check.json` with a console transcript in `data/cross_check.txt`.

13.3 Actual reachable-state counts

The bound $(k + 1)m^{k+1}$ proves finiteness but is not generally tight. The straightforward construction, before language minimization and with no explicit rejecting sink, gives the following exact counts.

k	$m = 2$	$m = 3$	$m = 4$	$m = 5$
2	13	35	71	139
3	18	58	124	288
4	23	75	161	377
5	28	92	198	466
6	33	109	235	555
7	38	126	272	644
8	43	143	309	733

Table 8: Reachable states in the generic residue construction. These numbers are not the state counts of the separately derived parity quotient, and no formula for general m is inferred from the table.

13.4 Why a finite recurrence certificate proves an identity

Lemma 13.1 (Cayley–Hamilton certificate). *Suppose $u_n = \lambda A^n f$ for an $N \times N$ matrix A over $\mathbb{Q}$. Let $Q(z) = 1 + q_1 z + \cdots + q_d z^d$ and let $P(z)$ have degree less than d . To prove $\sum_{n \geq 0} u_n z^n = P(z)/Q(z)$, it suffices to check that $Q(z) \sum u_n z^n$ agrees with $P(z)$ through degree $d - 1$, and that its next N coefficients, in degrees $d, \dots, d + N - 1$, are zero.*

Proof. For $n \geq 0$, define

$$r_n = u_{n+d} + q_1 u_{n+d-1} + \cdots + q_d u_n = \lambda A^n (A^d + q_1 A^{d-1} + \cdots + q_d I) f.$$

By Cayley–Hamilton, r_n obeys the order- N recurrence supplied by the characteristic polynomial of A . Its first N values being zero forces all its values to be zero. The initial coefficient check then proves the formal identity $Q \sum u_n z^n = P$. $\square$

[Kept twice: proof-hygiene mechanisms.] Thus the state-space dimension, not an arbitrary choice of a large experimental cutoff, determines how many checks certify a candidate. The argument does not require the candidate denominator to be the full characteristic polynomial, and it permits substantial cancellation. The complementary hygiene statement for the signed route — that no rational generating function is assumed and no unproved word-rewriting monoid is used — is Appendix A.

13.5 Two explicit modulo-three identities

For the following formulas the residue is one, so internal zeros are irrelevant. Define

$$\begin{aligned} P_2(z) &= 1 - 2z + 4z^2 - 6z^3 + 8z^4 - 10z^5 + 8z^6 - 6z^7, \\ Q_2(z) &= 1 - 4z + 8z^2 - 12z^3 + 16z^4 - 20z^5 + 19z^6 - 12z^7 + 4z^8, \end{aligned}$$

and

$$\begin{aligned} P_3(z) &= 1 - 2z + 4z^3 - 6z^4 + 8z^6 - 10z^7 + 8z^9 - 6z^{10}, \\ Q_3(z) &= 1 - 4z + 4z^2 + 4z^3 - 12z^4 + 8z^5 + 8z^6 - 20z^7 \\ &\quad + 11z^8 + 8z^9 - 12z^{10} + 4z^{11}. \end{aligned}$$

Then the finite certificates establish

$$H_{3,1}^{(2)}(z) = \frac{P_2(z)}{Q_2(z)}, \quad H_{3,1}^{(3)}(z) = \frac{P_3(z)}{Q_3(z)}. \quad (13.1)$$

These expressions appear in Stanley’s discussion [1, p. 24]; their forms are not claimed as discoveries here. The $k = 2$ congruence case is already covered by his earlier theorem.

For $k = 2$, the constructed matrix has dimension 35 and $\deg Q_2 = 8$. The certificate checks the eight initial numerator coefficients and 35 consecutive zero residuals. For $k = 3$, the corresponding numbers are 58, 11, and 58. The JSON files contain every reachable counter state, every labeled transition, the numerator and denominator arrays in ascending coefficient order, the exact sequence prefix, and all residuals. Rebuilding the graph and checking these integers reproduces the all- n proof via Lemma 13.1. This part is a finite computational certificate; the symbolic proof for arbitrary k and modulus remains separate.

14 Scope, limitations, and the next mathematical questions

14.1 What the argument establishes

The finite-state construction addresses the coefficient-residue conjecture for the entire family of products (1.2), not just several numerical values of k or prime moduli. It provides a proof mechanism, an explicit algorithm

for the rational function, a closed parity formula, and a rationality result for fixed exact multiplicities. The normal-form, fiber-factorization, and counter arguments are all given in full.

The signed construction addresses the parity statement alone, but does so for every quasifibonacci weight sequence with superincreasing seeds and for all 2^k coherent signings, and yields in addition signed flatness, the infinite-product corollary, the positive recurrence, the finite multinomial formula, minimality of the recurrence order, and the large-order expansion (11.7).

The distinction between established background and the proposed contribution is important. General finite-state rationality is a standard linear-algebra fact. Carry-block shapes have a precedent in Stanley's work. Flatness of truncated all-minus Fibonacci products at $k = 2$ is Zhao's [4], and the quasifibonacci seed conditions together with the even-order infinite all-minus theorem are Diao's [3]. The present routes combine a unique canonical cross-section with an exact fiber counter, and a periodic coherent signing with a pointwise two-tail cancellation. The supplied evidence supports those routes; it does not, by itself, certify that no equivalent proof exists elsewhere.

14.2 What is not being claimed

The argument is not a proof of every conjecture in Stanley's paper. In particular, it does not automatically cover arbitrary initial polynomials, arbitrary lists of factor coefficients, products with several nonconstant monomials per factor, or arbitrary sequences satisfying a constant-coefficient recurrence. Such changes may destroy either the regular cross-section or the finite counter used here. No claim is made that the generic residue state counts in Table 8 are minimal, or that every denominator has an unexplained universal form. No higher-moment conjecture from the same paper is claimed solved. Coherent signing is proved sufficient for flatness; *all* flat signings are not classified. No explicit bijection between the colored compositions of Section 10 and odd-coefficient positions is supplied; their equinumerosity is proved by generating functions only.

[Kept twice: initialization boundaries.] The restriction $k \geq 2$ is genuine. At $k = 1$, all weights equal one and $P_n(x) = (1 + x)^n$; the canonical-injectivity argument no longer applies. Even the displayed parity formula is already wrong at $n = 4$: it would predict four odd coefficients, whereas $(1 + x)^4$ has two. This boundary case is excluded throughout, as in the nontrivial $k \geq 2$ framework of the source's pair construction. It is a different degeneration from the failure of non-superincreasing seeds recorded in Remark 9.13.

For moduli greater than two, changing a factor's sign no longer leaves its residue unchanged, so the parity lift (9.14) is unavailable in that form. The signed route therefore does *not* prove Theorem 1.1; that theorem rests entirely on the finite-state route. Conversely, the finite-state route is nailed to the initialization (2.1): its canonical normal form and carry blocks are derived from exactly that initial segment, and it does not prove Theorem 1.4.

The manuscript has not been independently refereed or formalized in a proof assistant. The computational checks are exact but finite; their role is not to substitute for the general proofs. The special Cayley–Hamilton certificates are distinguished because a proved finite dimension bound makes those finite checks logically sufficient. The numerical constants are high-precision approximations, not certified interval enclosures.

14.3 Concrete continuations

One natural next problem is to classify the reachable modulo- m counter vectors uniformly in k , as was done here for $m = 2$. A sufficiently explicit classification for $m = 3$ might explain the relation between the two displayed fractions in (13.1) and give a general symbolic formula. The state-count table suggests structure, but is not offered as a theorem outside its checked range.

A second question is whether a finite collection of root-of-unity-weighted products can replace the parity lift at moduli greater than two. Flatness over $\{-1, 0, 1\}$ by itself supplies no such result.

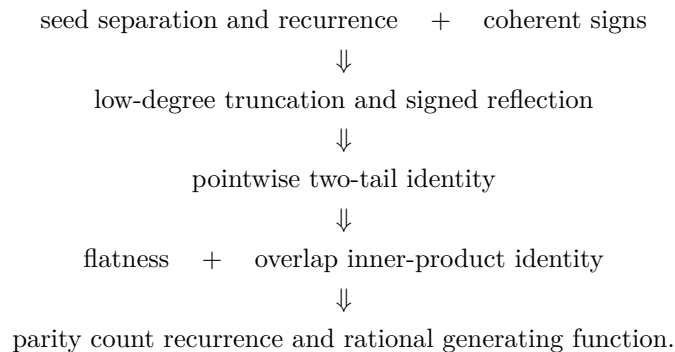
A third question is how much of the fiber-counting method survives for other numeration systems. Proposition 6.1 isolates the necessary ingredients: a regular set of unique representatives and a finite-dimensional nonnegative counter. Establishing either ingredient for a new recurrence is a substantive mathematical problem, rather than an automatic consequence of the recurrence being linear.

A fourth is which weaker seed conditions preserve the same count. The signed proof uses the seed condition twice — to obtain the initial values 2^n , and to establish the separation inequalities of Lemma 9.4 that make the tails disjoint. Changing only the initial values without checking that separation would not justify reusing the argument.

Finally, a formal proof could focus on the short dependency chain of the finite-state route: termination and canonical injectivity; the rewrite-count identity and its block decomposition; the two counter maps; and the finite-state resolvent. The uniform parity table and its elimination would then form a separate, comparatively small algebraic verification. The signed route’s chain, displayed in Appendix A, is shorter still and uses only elementary arithmetic.

A A compact proof dependency map for the signed route

The logical order is



Concretely: Lemma 9.4 and Lemma 9.7 feed (9.12) and Lemma 9.8; these give Proposition 9.9; that gives Theorem 9.3(i)–(ii) and Lemma 9.12; and those give Theorem 9.3(iii). Five lemmas and one proposition, all elementary arithmetic.

In particular, neither an assumed rational generating function nor any unproved description of a word-rewriting monoid is used in this argument. All boundary cases are covered by the convention that coefficient functions vanish outside their finite ranges. This is the signed route’s counterpart to Lemma 13.1: together they are the two answers this manuscript gives to the question of whether anything here is numerical curve fitting.

B Reproduction guide and artifact map

The archive is self-contained except for a Python interpreter, a $\text{\TeX}$ distribution for rebuilding the PDF, and the optional symbolic-check package. Python 3.10 or newer is required; no third-party package is needed except for `-symbolic`. From the archive root, run

```
python code/verify.py
python code/verify.py --symbolic
python code/check_certificates.py
python code/verify_signed.py
python code/test_api.py
python code/cross_check.py
python code/growth_constants.py
python code/make_tex_tables.py
python code/kbonacci_residues.py 3 5 20
python code/kbonacci_parity.py 3 8 --coefficient 105
```

The ninth command prints, for each $n = 0, \dots, 20$, the five residue counts for $k = 3, m = 5$, including internal zero positions; add `-exclude-holes` to count nonzero terms only. The tenth returns the JSON object with

parity equal to 1 and `signed_coefficient` equal to -1 . Do not run `verify_signed.py` under `python -O`: it refuses to start when assertions are disabled. `test_api.py` is unaffected, because `unittest` assertions are method calls that `-O` does not remove; `verify.py`, `check_certificates.py` and `cross_check.py` use explicit error checks and are safe under `-O`.

The independent script `check_certificates.py` validates the two modulo-three certificate files without importing the main implementation.

Reusable functions of `kbonacci_residues.py` include `weights`, `expand`, `normalize`, `build_machine`, `residue_counts`, `parity_machine`, `dfa_sequence` and `odd_counts`; `coefficient_for_canonical` implements the exact counter. For exact multiplicity q , call `build_machine(k, cap=q+1)` and take column q of its distributions. Words are always passed in low-to-high weight order.

The signed route's module is `kbonacci_parity.py`. It supplies `standard_seeds`, `check_seeds`, `weights` and `coherent_signs` for setting up a weight and sign sequence; `count_prefix` and `count_fast` for counting; `parity_bitset` and `signed_sparse` for direct expansion; and the class `CoefficientOracle`, whose `signed` and `parity` methods are described in Section 12.2. Documented invariants used as regression tests include

```
count_fast(3, 20) == 76576
count_fast(3, 10**12, modulus=1000000007) == 699562826
coefficient_for_canonical((0,0,0,0,1), k=2) == 3
normalize((1,1,0), k=2) == (0,0,1)
```

all of which `code/cross_check.py` asserts.

Artifact	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Editable manuscript and compiled article.
<code>code/kbonacci_residues.py</code>	Finite-state constructions and a command-line interface.
<code>code/kbonacci_parity.py</code>	Signed-cancellation algorithms and a command-line interface.
<code>code/verify.py</code>	Finite-state tests and the finite certificates.
<code>code/verify_signed.py</code>	Signed-route tests and numerical data generation.
<code>code/cross_check.py</code>	Comparisons between the two implementations.
<code>code/test_api.py</code>	Boundary cases and input validation.
<code>code/check_certificates.py</code>	Independent certificate validator.
<code>code/make_tex_tables.py</code>	Regenerates the three <code>data/*_table.tex</code> fragments.
<code>code/growth_constants.py</code>	Floating-point growth table of Table 4.
<code>data/coefficient_counts.csv</code>	All 1,666 directly checked residue counts.
<code>data/odd_counts.csv</code>	Odd counts through length 100 for $2 \leq k \leq 8$.
<code>data/counts.csv</code>	Odd counts through length 200 for $2 \leq k \leq 20$.
<code>data/state_counts.csv</code>	Reachable-state counts in Table 8.
<code>data/growth.csv</code>	80-digit growth constants, Table 3.
<code>data/growth_constants.csv</code>	Floating-point growth constants, Table 4.
<code>data/large_values.json</code>	Complete $a_k(10000)$ and large-index modular values.
<code>data/b_k*.txt</code>	Two-column index/value files; not assigned OEIS b-files.
<code>data/verification_summary.json</code>	Finite-state verification ranges and results.
<code>data/verification_signed.json</code>	Signed-route verification ranges and results.
<code>data/cross_check.json</code>	Outcome of the merged cross-verification.
<code>data/initial_table.tex</code>	Generated fragments required to compile the PDF.
<code>data/growth_table.tex</code>	
<code>data/large_table.tex</code>	
<code>certificates/mod3_k2.json</code>	Complete finite certificates for (13.1).
<code>certificates/mod3_k3.json</code>	
<code>certificates/parity_k*.json</code>	
<code>README.md</code>	Usage instructions for the merged archive.
<code>LITERATURE.md</code> , <code>RESEARCH_STATUS.md</code>	The two literature and status audits.

Compile the article with `latexmk -pdf article.tex`, or with `pdflatex article.tex` twice, or use the provided Makefile. The three generated fragments in `data/` must exist; the eighth command above regenerates them. Numerical growth approximations are regenerated by the seventh. Those floating-point values are illustrative and do not enter any exact verification certificate. The archive contains no third-party research PDFs, font files, compiled Python caches, or L^AT_EX build intermediates.

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